

Evidence-Graded Agentic Retrieval-Augmented Generation for Trustworthy Document Intelligence over Visually Rich Business Documents

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BY
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Declaration of Originality

I, Student Name, with SR No. SR Number hereby declare that the material presented in the project report titled

Evidence-Graded Agentic Retrieval-Augmented Generation for Trustworthy Document Intelligence over Visually Rich Business Documents

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I also thank my peers, reviewers, and colleagues for their discussions on practical document processing problems, user expectations, and the need for reliable evidence-backed answers. These discussions influenced the design of the validation layer, confidence reporting, and Agentic RAG workflow.

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Abstract

Visually rich business documents such as contracts, invoices, reports, and scanned forms remain difficult for automated systems because their information is distributed across text, layout, tables, page structure, and domain-specific field conventions. Recent document AI models improve extraction and document understanding, while Retrieval-Augmented Generation (RAG) improves access to external knowledge. However, practical document question answering still suffers from three research problems: extraction uncertainty propagates into retrieval, retrieved evidence may be only partially relevant, and language models can generate unsupported or contradictory answers even when retrieval is used.

This thesis investigates the research question of how uncertainty-aware document extraction and evidence-graded Agentic RAG can improve trustworthy document intelligence for semi-structured and visually rich business documents. The implemented system is used as a research prototype to study an end-to-end pipeline that combines OCR, layout-aware document analysis, structured field extraction, validation, semantic retrieval, evidence grading, and grounded generation. Unlike a direct OCR-to-RAG baseline, the proposed workflow introduces query planning, structured-field shortcutting, candidate evidence retrieval, evidence scoring, confidence estimation, and traceable answer generation.

The central hypothesis is that document question answering becomes more reliable when the system explicitly separates four decisions: whether a question can be answered from structured extractions, which chunks should be retrieved, which retrieved evidence is sufficiently relevant, and whether the final answer is grounded enough to return. This framing is aligned with recent research on practical document understanding benchmarks, RAG evaluation, groundedness, hallucination detection, and self-reflective retrieval-augmented generation.

A preliminary prototype evaluation reports a field extraction F1 score of 0.8, Search Precision@5 of 0.25, Search Mean Reciprocal Rank of 1.0, estimated throughput of 29.77 documents per hour, and a review rate of 0.152. These early results are treated as pilot evidence rather than final claims. The thesis further defines a research-oriented evaluation protocol comparing multiple baselines: OCR-only extraction, OCR plus direct RAG, structured extraction plus direct RAG, and the proposed evidence-graded Agentic RAG workflow. The proposed metrics include extraction F1, Precision@k, MRR, answer faithfulness, context precision, context recall, unsupported-claim rate, latency, and human-review load.

The work contributes a reproducible prototype, a research framework for uncertainty-aware document QA, a baseline comparison plan, and a PhD-ready research agenda for trustworthy multimodal document intelligence. Future work will expand the benchmark dataset, add labelled question-answer pairs, introduce stronger table and layout modelling, fine-tune document understanding models, and evaluate faithfulness and hallucination behaviour at scale.

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Introduction

Business and institutional workflows depend heavily on visually rich documents: invoices, contracts, purchase orders, reports, scanned forms, and supporting records. These documents contain information in text, layout, tables, headers, footers, signatures, and page ordering. Although large language models can answer natural-language questions, and OCR can convert images into text, neither component alone is sufficient for trustworthy document intelligence. The central challenge is not only reading a document, but deciding which extracted information is reliable, which evidence supports an answer, and when the system should decline or request human review.

Recent research in Document AI has introduced multimodal models such as LayoutLMv3 for text-layout-image understanding [5], OCR-free document understanding approaches such as Donut [6], and more practical benchmarks for visually rich documents such as DUDE [7]. At the same time, RAG has become a dominant pattern for grounding language model outputs in external knowledge [8]. However, RAG does not eliminate hallucination by itself. Retrieved evidence can be irrelevant, incomplete, or contradictory, and generated answers can still include unsupported claims [11].

Research Motivation

Most practical document intelligence systems are evaluated as engineering pipelines: upload a document, extract fields, index text, and answer questions. For an M.Tech project this is useful, but for a research-oriented thesis the deeper question is whether the pipeline improves trustworthiness in a measurable way. In document-heavy domains, an answer is useful only if it is traceable to reliable evidence. A wrong amount, an unsupported contract interpretation, or an invented date can be more damaging than no answer.

This project therefore reframes the implemented application as a research prototype for uncertainty-aware and evidence-graded document question answering. The goal is to study whether explicit intermediate decisions can improve reliability: field extraction confidence, validation status, retrieval relevance, evidence grading, answer confidence, and human-review routing.

Research Gap

The literature suggests three gaps that motivate this thesis:

1. Extraction-to-answer uncertainty gap: Document AI systems often report extraction confidence, while RAG systems focus on retrieval and generation. There is limited integration between extraction uncertainty and downstream question-answer confidence.
2. Evidence selection gap: Direct RAG pipelines retrieve chunks but do not always explicitly grade whether the retrieved context is sufficient for the question. This is problematic for long, multi-page, visually structured documents.
3. Evaluation gap: RAG systems are difficult to evaluate because retrieval, generation, faithfulness, and answer relevance are separate dimensions [12]. Document intelligence systems additionally need extraction accuracy, review rate, and layout-sensitive benchmarks.

Research Questions

The thesis is structured around the following research questions:

1. RQ1: How can OCR, layout-aware document understanding, rule-based extraction, and validation be combined to produce uncertainty-aware document representations?
2. RQ2: Does structured-field lookup improve exact-field question answering compared with direct RAG over extracted text?
3. RQ3: Does evidence grading improve the reliability of RAG answers by filtering weak or irrelevant retrieved chunks before generation?
4. RQ4: What evaluation protocol can jointly measure extraction quality, retrieval quality, answer faithfulness, latency, and human-review load for document intelligence?

Hypotheses

The research questions lead to the following hypotheses:

1. H1: A hybrid extraction pipeline combining digital text extraction, OCR, layout-aware classification, and validation will produce more useful document representations than OCR text alone.
2. H2: For exact factual questions, structured-field lookup will reduce latency and unsupported generation compared with direct RAG.
3. H3: Evidence grading before generation will improve answer faithfulness and reduce unsupported claims compared with a direct retrieve-and-generate baseline.
4. H4: A multi-metric evaluation protocol will reveal trade-offs that single aggregate scores miss, especially between accuracy, retrieval quality, latency, and review effort.

Proposed Research Contribution

The proposed contribution is an evidence-graded Agentic RAG framework for trustworthy document intelligence. The framework uses a modular pipeline that first creates an uncertainty-aware document representation, then performs query planning, structured lookup, evidence retrieval, evidence grading, and grounded answer generation. The framework is not presented as a final solved system; rather, it is a research scaffold for studying reliability, uncertainty propagation, and evaluation in document-grounded question answering.

The expected research contributions are:

1. A reproducible full-stack prototype for document intelligence over uploaded business documents.
2. An uncertainty-aware extraction and validation layer that stores confidence, review flags, and processing logs.
3. An Agentic RAG workflow that separates query planning, structured lookup, retrieval, evidence grading, and generation.
4. A baseline comparison design covering OCR-only, direct RAG, structured extraction plus RAG, and evidence-graded Agentic RAG.
5. A PhD-oriented research agenda for trustworthy multimodal document intelligence.

Scope and Assumptions

The current system is a local research prototype. It supports PDF and image-based document inputs, stores metadata and extracted fields in SQLite, and serves a React frontend through a FastAPI backend. The prototype uses available OCR, embedding, and language-model components rather than training all models from scratch. This is appropriate for the present thesis because the research focus is integration, uncertainty-aware orchestration, and evaluation design.

The current benchmark is small and should be interpreted as pilot evidence. For a PhD-level continuation, the dataset should be expanded with labelled fields, document-level labels, retrieval relevance judgments, and question-answer pairs. The proposed future work section defines this path explicitly.

Organisation of the Report

The remainder of this report is organised as follows. Chapter 2 reviews Document AI, OCR, layout-aware understanding, RAG, hallucination, and evaluation literature. Chapter 3 presents the proposed evidence-graded Agentic RAG framework and implementation. Chapter 4 describes pilot results and a research-grade evaluation protocol. Chapter 5 summarises the thesis and proposes a PhD research agenda.

Literature Review

Document AI and Visually Rich Documents

Intelligent Document Processing has evolved from OCR-based text extraction into multimodal document understanding. Visually rich documents contain textual content, spatial layout, tables, logos, signatures, section hierarchy, and page-level structure. Therefore, document understanding requires more than converting images into plain text.

LayoutLMv3 represents a major direction in multimodal document AI. It pre-trains document representations using unified text and image masking and learns alignment between words and image patches [5]. Such models are useful because key-value relations in documents are often spatial rather than purely sequential. For example, an amount near a "Total" label has a different meaning from an amount inside a paragraph.

OCR-free models such as Donut challenge the classical OCR-first pipeline by directly mapping document images to structured outputs [6]. These approaches reduce dependence on OCR quality, but they usually require significant training data and compute. For a practical research prototype, the hybrid OCR plus layout-aware path remains useful because it can run locally, expose intermediate confidence values, and support human audit.

Practical Benchmarks for Document Understanding

Earlier document AI datasets often focused on narrow tasks such as receipts or forms. More recent benchmarks emphasise multi-domain, multi-page, and visually rich settings. The DUDE benchmark argues that practical document understanding needs evaluation scenarios closer to real-world documents, with diverse questions, answers, layouts, and domains [7]. This is important for the present thesis because the target documents are business documents rather than one fixed template.

A research-centric thesis should therefore not evaluate only whether the app runs. It should evaluate extraction accuracy, retrieval ranking, document QA, faithfulness, latency, and review burden across varied documents. This motivates the multi-metric protocol proposed in Chapter 4.

OCR and Extraction Uncertainty

OCR remains a foundational component in many document systems. Tesseract is a widely used open-source OCR engine [1], and EasyOCR provides ready-to-use OCR models across many scripts and languages [2]. However, OCR systems can fail on low-resolution scans, rotated pages, poor contrast, unusual fonts, handwritten content, and complex tables.

A key research issue is uncertainty propagation. OCR may provide word-level confidence, extraction rules may provide field-level confidence, and validation rules may flag suspicious values. Yet many RAG systems discard this uncertainty and pass all extracted text to a retriever. This thesis argues that document QA should preserve uncertainty signals and use them during answer generation and review routing.

Retrieval-Augmented Generation

RAG combines retrieved external context with a parametric language model [8]. It is useful for document intelligence because it allows answers to be grounded in uploaded documents instead of relying on model memory. However, RAG is not a guarantee of factuality. Retrieval may miss relevant evidence, retrieve only partial context, or surface misleading text. The generator can also introduce unsupported claims.

Self-RAG extends this direction by introducing self-reflection over retrieval and generation decisions [9]. The idea that a system should decide when to retrieve, critique retrieved passages, and critique its own output is relevant to the present work. The proposed Agentic RAG workflow is simpler and engineering-oriented, but follows a similar research motivation: expose intermediate decisions and reduce unsupported answers.

Groundedness and Hallucination in RAG

Recent studies show that groundedness remains an open problem in retrieval-augmented long-form generation [10]. Even with retrieved context, generated answers may include claims that are unsupported by evidence. RAGTruth further shows the value of hallucination corpora for trustworthy retrieval-augmented language models, including word-level annotations of unsupported claims [11].

These works motivate evidence grading and answer faithfulness metrics in this thesis. A document QA system should not only retrieve chunks; it should estimate whether the chunks actually support the requested answer. If the evidence is weak, the system should respond with insufficient evidence rather than generate a confident answer.

RAG Evaluation

RAG evaluation is multi-dimensional. RAGAS proposes reference-free metrics for evaluating retrieval-augmented generation, including dimensions related to context relevance, answer faithfulness, and answer relevance [12]. This is useful because labelled answers are expensive to create, especially for domain-specific documents.

For this thesis, RAG evaluation is combined with document-specific metrics. Extraction F1 measures whether structured fields are correctly extracted. Precision@k and MRR measure retrieval ranking. Faithfulness and unsupported-claim rate measure whether generated answers stay grounded. Review rate measures how often the system requires human verification. Latency and throughput measure operational feasibility.

Research Positioning

The proposed work is positioned between three research areas: Document AI, trustworthy RAG, and human-in-the-loop validation. It does not claim to introduce a new transformer architecture. Instead, its research novelty lies in the orchestration and evaluation of a document-grounded QA pipeline that preserves uncertainty signals, routes exact questions through structured fields, grades evidence before generation, and exposes traceable outputs.

This positioning is appropriate for a PhD proposal because it opens several research extensions: uncertainty propagation from OCR to RAG, multimodal retrieval over text and layout, evidence sufficiency

estimation, hallucination-aware answer generation, and benchmark construction for document-grounded QA.

Summary

The literature indicates that trustworthy document intelligence remains an active research area. OCR, layout-aware models, RAG, and LLMs each solve part of the problem, but the end-to-end reliability question remains unresolved. This thesis addresses that gap by proposing and prototyping an evidence-graded Agentic RAG framework for visually rich business documents.

Methodology and Research Design

Overview of the Proposed Framework

The proposed framework is an evidence-graded Agentic RAG pipeline for trustworthy document intelligence. The system is implemented as a full-stack prototype, but the methodology is framed as a research design. The pipeline has two stages: document representation and question answering. The document representation stage creates an uncertainty-aware document store. The question-answering stage uses query planning, structured lookup, retrieval, evidence grading, and grounded generation.

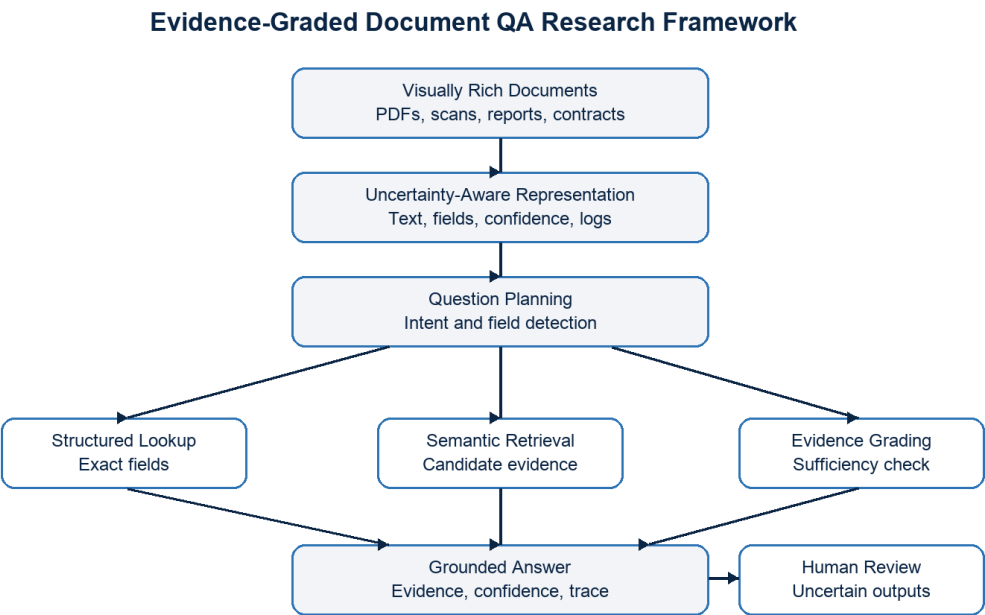


Figure 1: High-level architecture of the document intelligence system

Research Prototype

The prototype uses a React frontend and FastAPI backend. SQLite stores documents, extracted fields, confidence values, review flags, full text, processing logs, and chat history. The backend exposes API endpoints for upload, reprocessing, validation, search, statistics, evaluation, and Agentic RAG. The implemented system allows research experiments to be run against a real application rather than isolated notebooks.

Document Representation Stage

The first stage transforms an uploaded file into an uncertainty-aware representation. Digital PDFs are processed using embedded text extraction. Scanned documents are processed using OCR. LayoutLMv3 is used for document type classification and layout-aware feature extraction. Rule-based and LLM-assisted extraction identify structured fields. Validation rules then flag suspicious values.

The important research decision is to store not only final text, but also metadata that can influence downstream trust: document type, extraction confidence, source, page number, validation status, review flag, processing time, and logs. This enables later experiments on whether downstream QA improves when these uncertainty signals are used.

Uncertainty-Aware Document Representation

1. Receive uploaded document and create metadata record
2. Detect digital PDF or scanned-document path
3. If digital text is available
4. Extract embedded text directly
5. Classify first page using layout-aware model
6. Else
7. Convert pages to images
8. Extract OCR text and word confidence
9. Classify pages and document type using layout-aware model
10. End if
11. Merge page text and table text
12. Extract structured fields using rules and optional LLM extraction
13. Store field confidence and source
14. Validate fields using domain rules
15. Mark uncertain fields for review
16. Index document chunks for retrieval

Question Answering Stage

The second stage answers user questions. A direct RAG system would retrieve chunks and immediately ask a language model to generate an answer. The proposed workflow inserts three additional research mechanisms: intent-aware query planning, structured-field lookup, and evidence grading.

Query planning classifies the question into intents such as field lookup, summary, comparison, risk review, or grounded QA. Structured-field lookup is used when the question asks for an exact extracted field such as invoice number, contract parties, amount, or date. If structured lookup fails, the system retrieves semantic chunks, grades evidence, and only then asks the language model to answer.

Evidence-Graded Agentic RAG

1. Receive question and optional document filter
2. Detect query intent and possible target field
3. If question can be answered from structured fields
4. Return structured answer with extraction confidence and evidence
5. Else
6. Retrieve candidate chunks using semantic search
7. Score each evidence item using similarity, term coverage, and intent match
8. Select top evidence items above sufficiency threshold
9. If evidence is insufficient
10. Return insufficient-evidence response

11. Else
12. Generate answer using selected evidence only
13. Return answer, evidence, confidence, trace, and latency
14. End if
15. End if

Baseline Systems

A research-centric evaluation requires baselines. The following baselines are defined for comparison:

1. B0: OCR-only extraction: extract text and apply rule-based fields without semantic retrieval.
2. B1: Direct OCR plus RAG: retrieve chunks from OCR/full text and generate an answer without structured lookup or evidence grading.
3. B2: Structured extraction plus direct RAG: use structured fields for display, but answer questions through direct RAG.
4. B3: Proposed evidence-graded Agentic RAG: use planning, structured lookup, retrieval, evidence grading, and sufficiency thresholding.

Comparing these baselines allows the thesis to test the value of each added component rather than only reporting final application behaviour.

Variables and Metrics

The independent variable is the QA strategy: OCR-only, direct RAG, structured extraction plus RAG, or evidence-graded Agentic RAG. Dependent variables include extraction F1, retrieval precision, answer faithfulness, unsupported-claim rate, latency, throughput, and review rate.

Metric	Purpose	Research Question
Extraction F1	Measures key-field extraction quality	RQ1
Precision@5	Measures retrieval ranking quality	RQ3
MRR	Measures first relevant result rank	RQ3
Faithfulness	Measures answer support in evidence	RQ3, RQ4
Unsupported-claim rate	Measures hallucination risk	RQ3, RQ4
Latency	Measures user-facing efficiency	RQ2, RQ4
Review rate	Measures human verification burden	RQ1, RQ4

Research metrics for trustworthy document intelligence

System Implementation

The backend implements modular services for OCR, layout-aware classification, fusion, entity extraction, validation, search, semantic retrieval, RAG, Agentic RAG, and evaluation. The frontend provides pages for dashboard monitoring, upload, document review, search, and Agentic RAG. The application can serve the built frontend from FastAPI, making it a single runnable prototype for demonstration and evaluation.

Threats to Validity

The main internal validity threat is that current heuristics may overfit to the small pilot dataset. The main external validity threat is that business documents vary widely across domains. Construct validity may be affected if automatic faithfulness metrics do not fully match human judgement. To reduce these threats, future work should use a larger document corpus, multiple annotators, held-out document templates, and both automatic and human evaluation.

Reproducibility

The research prototype is organised into separate frontend, backend, service, model, and evaluation modules. The evaluation endpoint exposes computed metrics, and the API design allows repeated testing over uploaded documents. For a PhD continuation, the benchmark data, prompts, retrieved evidence, generated answers, and human labels should be stored as versioned experiment artifacts.

Pilot Results and Research Evaluation Plan

Purpose of Evaluation

The purpose of evaluation is to determine whether the proposed evidence-graded Agentic RAG framework improves trustworthy document intelligence compared with simpler baselines. The current prototype provides pilot metrics, while the full research plan defines the experiments required for a PhD-level study.

Pilot Prototype Setup

The prototype was evaluated in a local environment using the FastAPI backend, React frontend, SQLite database, OCR services, document classification service, extraction layer, validation layer, search layer, and Agentic RAG endpoint. The application statistics endpoint reported five uploaded documents, two completed documents, two documents requiring review, and an average processing time of 120941 ms.

The current benchmark file was available through the evaluation endpoint. The measured values should be interpreted as early pilot results because the benchmark is small and not yet representative of a full research dataset.

Pilot Metrics

The preliminary metrics are shown in Table 2.

Metric	Pilot Value	Interpretation
Field Extraction F1	0.8	Promising but needs larger benchmark
Search Precision@5	0.25	Top-five ranking needs improvement
Search MRR	1.0	First relevant result appeared early
QA Answer Accuracy	Not measured	Requires labelled QA benchmark
Throughput	29.77 docs/hr	Suitable for local prototype
Review Rate	0.152	Indicates human-review routing is active

Pilot evaluation metrics from the current prototype

Interpretation of Pilot Results

The Field Extraction F1 score of 0.8 suggests that rule-based and assisted extraction can capture useful structured fields in the pilot dataset. However, a larger dataset is required before making general claims.

Search MRR of 1.0 indicates that the first relevant result appeared at the top for the benchmark cases, while Precision@5 of 0.25 suggests that the broader ranked list includes irrelevant results. This points to a clear research opportunity in hybrid retrieval, reranking, and evidence grading.

The review rate of 0.152 indicates that the system is not blindly accepting every extracted value. From a trustworthiness perspective, this is valuable because uncertain fields are surfaced for human review. From an operational perspective, excessive review load can reduce automation benefits. Therefore, review rate should be studied as a trade-off against extraction F1 and answer faithfulness.

Proposed Research Dataset

A PhD-ready evaluation should use a larger dataset with four annotation layers:

1. Document labels: document type, page count, scan quality, and source.
2. Field labels: ground-truth values for dates, amounts, parties, invoice numbers, purchase orders, and other domain fields.
3. Retrieval labels: relevance judgments for chunks retrieved for each query.
4. QA labels: ground-truth answers, acceptable answer variants, and supporting evidence spans.

Baseline Comparison Protocol

The main experiment should compare four systems:

1. B0: OCR-only extraction.
2. B1: Direct OCR plus RAG.
3. B2: Structured extraction plus direct RAG.
4. B3: Evidence-graded Agentic RAG.

Each system should be evaluated over identical documents and questions. Exact-field questions should test whether structured lookup improves correctness and latency. Summary, comparison, and risk-review questions should test whether evidence grading improves faithfulness and reduces unsupported claims.

Evaluation Metrics

The full evaluation protocol should include:

1. Extraction metrics: precision, recall, F1, field-level confidence calibration, and validation error rate.
2. Retrieval metrics: Precision@1, Precision@5, MRR, context precision, and context recall.
3. Generation metrics: answer correctness, faithfulness, answer relevance, unsupported-claim rate, citation accuracy, and refusal appropriateness.
4. Operational metrics: latency, throughput, memory usage, and human-review rate.
5. Human evaluation: expert judgement of answer usefulness, evidence sufficiency, and trust.

Expected Comparative Analysis

The expected outcome is not that the proposed system dominates every metric. Instead, the research should identify trade-offs. OCR-only extraction is expected to be fast but limited for question answering. Direct RAG is expected to answer broad questions but may hallucinate or cite weak evidence. Structured

extraction plus direct RAG should improve exact questions but may not fully solve faithfulness. Evidence-graded Agentic RAG is expected to improve traceability and reduce unsupported answers, possibly at the cost of additional latency.

Ablation Studies

The following ablation studies are proposed:

1. Remove structured-field lookup and measure exact-field QA accuracy and latency.
2. Remove evidence grading and measure faithfulness and unsupported-claim rate.
3. Remove validation flags and measure downstream answer confidence calibration.
4. Compare keyword retrieval, dense retrieval, and hybrid retrieval.
5. Compare different confidence thresholds for review routing.

Statistical Analysis

For a larger dataset, paired comparisons should be used because each system answers the same question set. Retrieval and QA metrics can be compared using paired significance tests or bootstrap confidence intervals. Inter-annotator agreement should be reported for human-labelled relevance and faithfulness judgments.

Discussion

The pilot results show that the implemented prototype is a viable platform for research. The current extraction and retrieval metrics are sufficient to justify deeper experimentation, but not sufficient for final research claims. The strongest PhD path is therefore to treat this M.Tech work as a prototype and convert it into a rigorous study on uncertainty-aware, evidence-graded document QA.

Summary

The current system provides measurable pilot evidence and a working end-to-end implementation. The research evaluation plan extends this into a publishable experimental design with baselines, ablations, multiple metrics, and human evaluation.

Conclusion and PhD Research Agenda

Conclusion

This thesis presents a research-oriented document intelligence prototype for visually rich business documents. The work moves beyond a simple application implementation by framing the system as an evidence-graded Agentic RAG framework. The framework combines uncertainty-aware document extraction, validation, structured lookup, semantic retrieval, evidence grading, and grounded generation.

The central research claim is that trustworthy document question answering requires explicit intermediate decisions. A system should know when to answer from structured fields, when to retrieve context, whether retrieved context is sufficient, and when to ask for human review. This is particularly important for business documents where wrong answers can affect financial, contractual, or operational decisions.

The current pilot results show that the prototype is measurable and functional. The field extraction F1 score of 0.8, Search MRR of 1.0, throughput of 29.77 documents per hour, and review rate of 0.152 demonstrate that the system has reached a useful proof-of-concept stage. At the same time, the lower Search Precision@5 and missing QA benchmark show that the work is not yet complete as a research contribution. This honest gap is valuable because it defines the next research steps.

Research Contributions

The thesis contributes:

1. A working full-stack prototype for document-grounded extraction, search, and QA.
2. An uncertainty-aware representation layer that preserves confidence, validation, review, and trace information.
3. An evidence-graded Agentic RAG workflow that separates planning, structured lookup, retrieval, evidence scoring, and answer generation.
4. A baseline comparison plan for evaluating whether each system component improves trustworthiness.
5. A PhD-ready research agenda for trustworthy multimodal document intelligence.

Why This Can Support PhD Applications

A PhD proposal is usually evaluated on research potential rather than only implementation effort. This project can support PhD applications because it has a practical prototype, a clear research gap, measurable preliminary results, and a natural path to publishable experiments. It connects to active research areas: Document AI, RAG evaluation, hallucination detection, multimodal retrieval, human-in-the-loop validation, and trustworthy AI.

The most compelling PhD positioning is:

Trustworthy document intelligence requires uncertainty-aware extraction and evidence-graded generation, especially for visually rich business documents where answer correctness must be traceable to document evidence.

Limitations

The current work has several limitations. The benchmark dataset is small. The QA evaluation is not yet labelled. The document understanding model is not fine-tuned on project-specific data. Table extraction and layout modelling can be improved. The system currently runs locally and does not yet include production-grade authentication, access control, audit governance, or scalable vector storage.

These limitations should not be hidden. They should be presented as the motivation for future research. A strong PhD proposal often begins with a working prototype and then identifies the deeper unresolved research questions.

PhD Research Agenda

The following research agenda can extend this M.Tech work into a PhD proposal:

1. Dataset construction: Build a labelled benchmark of visually rich business documents with fields, evidence spans, retrieval relevance, and QA annotations.
2. Uncertainty propagation: Study how OCR confidence, extraction confidence, validation flags, and retrieval scores can be combined into answer-level confidence.
3. Evidence sufficiency modelling: Develop methods that decide whether retrieved evidence is enough to answer safely.
4. Multimodal retrieval: Retrieve not only text chunks but also layout regions, table cells, page positions, and document structure.
5. Faithfulness evaluation: Measure unsupported claims and citation correctness using both automatic metrics and expert human evaluation.
6. Human-in-the-loop learning: Use reviewer corrections to improve extraction, validation, retrieval, and answer generation over time.
7. Deployment and governance: Study privacy, access control, auditability, and compliance for enterprise document intelligence.

Potential Publication Directions

The work can be shaped into several publication directions:

1. A systems paper on evidence-graded Agentic RAG for document intelligence.
2. An evaluation paper on benchmark design for visually rich business document QA.
3. A methods paper on uncertainty propagation from OCR and extraction into RAG answer confidence.
4. An applied AI paper on human-in-the-loop validation for trustworthy document extraction.

Final Remarks

The current project is a strong foundation for an M.Tech thesis and can be reshaped into a credible PhD research proposal. Selection for a PhD program cannot be guaranteed, but the research-centric version strengthens the case by clearly identifying novelty, research questions, hypotheses, baselines, metrics, limitations, and future research directions. The next major step is to expand the dataset and run comparative experiments against the defined baselines.

API Endpoints

Method	Endpoint	Description
POST	/api/v1/documents/upload	Upload document for processing
GET	/api/v1/documents	List uploaded documents
GET	/api/v1/documents/{id}	Fetch document details and extractions
DELETE	/api/v1/documents/{id}	Delete document and related records
POST	/api/v1/documents/{id}/reprocess	Re-run document processing
GET	/api/v1/documents/{id}/validate	Validate extracted fields
POST	/api/v1/documents/{id}/ask	Ask a question about one document
POST	/api/v1/ask	Ask a question across all documents
POST	/api/v1/agent/ask	Ask through backend agent router
POST	/api/v1/agent-rag/ask	Ask using Agentic RAG workflow
POST	/api/v1/search	Search documents
GET	/api/v1/stats	Get processing statistics
GET	/api/v1/evaluation/metrics	Get evaluation metrics

Major backend API endpoints

Prototype Configuration

The prototype uses the following important configuration values:

1. API prefix: /api/v1
2. Local database: sqlite:///./doc_intelligence.db
3. Supported formats: PDF, PNG, JPG, JPEG, TIFF, and DOCX
4. Maximum file size: 50 MB
5. VLM model: microsoft/layoutlmv3-base
6. Embedding model: all-MiniLM-L6-v2
7. Confidence threshold: 0.7

Recommended Screenshots for Final Submission

The final submitted report should include screenshots of the following pages after personal details and final data are confirmed:

1. Dashboard page showing document counts and processing summary.
2. Upload page showing the document upload workflow.
3. Document detail page showing extracted fields and processing logs.
4. Search page showing document retrieval results.
5. Agentic RAG page showing answer, evidence snippets, confidence, and trace steps.

References

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